

FINAL CONCLUSIVE REPORT:

The computational refinement of the protein structure resulted in an overall improvement in several indicators of structural quality, although not all validation parameters improved. Comparison of the pre and post-refinement models showed that the ERRAT quality factor increased from 80.8728 to 97.7444, representing an improvement of 16.8716 points. This substantial increase indicates that the refined structure has considerably better overall non-bonded interaction quality and is therefore more structurally acceptable according to ERRAT analysis.

The favoured regions in the Ramachandran plot (PROCHECK) also increased from 82.5% before refinement to 95.2% after refinement, an improvement of 12.7 percentage points. At the same time, the proportion of allowed regions decreased from 16.6% to 4.8%, which is consistent with residues moving from the less-preferred allowed regions into the favoured regions. This represents a positive change in backbone dihedral-angle geometry and suggests that the refined model has improved stereochemical quality. The percentage of outliers remained unchanged at 0%, indicating that refinement did not introduce any residues into disallowed/outlier regions.

However, the Verify3D score decreased from 93.64% to 87.94%, corresponding to a 5.7 percentage-point decrease. Although this represents a decline, the post-refinement value remains relatively high, suggesting that the three-dimensional structure is still generally compatible with its amino-acid sequence.

Overall, the combined validation results indicate that computational refinement improved the structural quality of the selected protein model substantially. The major improvements were observed in ERRAT quality and PROCHECK stereochemical parameters, while outliers remained unchanged at zero. The only parameter that worsened was Verify3D, which showed a moderate decrease. Therefore, the refinement can be considered largely successful, because the magnitude of the improvements in ERRAT and Ramachandran favoured regions are far more than the moderate reduction in Verify3D performance.